Topic D: proofs and their costs

Human Compression project

The common paper on stabilizers, support and enumeration supports three complete applications: commuting involutions (D1), Suzuki polyhedra (D2), and Hall–Janko geometries (D3). The comparison preserves the thirty-seven admitted source clauses and their stated qualifications. The paired graph records actual proof routes; it does not replace the complete manuscripts by their shortest selected routes.

Written papers. The three complete source PDFs cost 47,894 characters. The whole common theory costs 15,632 and the three applications cost 29,855, giving

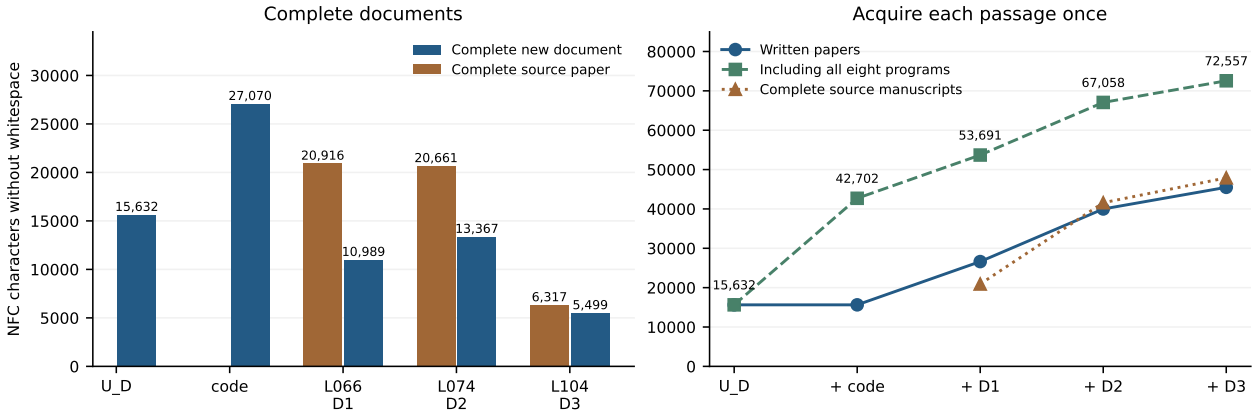
$$R_D^{\text{written}} = \frac{47\,894}{15\,632 + 29\,855} = \frac{47\,894}{45\,487} = 1.052916\ldots.$$

Title, abstract, definitions, proofs, tables, context and references are included. The unused normalizer theorem remains paid as part of the whole common paper.

Including the complete program. The twelve-page appendix preserves five required mathematical programs and three complete diagnostics. All eight are charged, including comments and the wrapper, at 27,070 characters. The separate comparison is

$$R_D^{\text{programs}} = \frac{47\,894}{45\,487 + 27\,070} = \frac{47\,894}{72\,557} = 0.660088\ldots.$$

The full eight raw files cost 26,185; substituting them for the printed appendix gives $47\,894/71\,672 = 0.668239\ldots$. They are alternatives, never additional charges. Two complete operational drivers cost 10,038 raw characters and 12,765 actual bytes, separately inventoried rather than included in these factors.



These figures use calibrated complete PDF extracted text, NFC characters without whitespace. All thirty policy coordinates are provided. Expanded external proofs, catalogue and algorithm construction, historical software, and graphics remain unknown costs, not zero.

After acquiring the whole common theory and all eight programs.

Source / application	Complete source	Complete application	Conditional factor
L066 / D1	20,916	10,989	1.903
L074 / D2	20,661	13,367	1.546
L104 / D3	6,317	5,499	1.149

Acquisition and reuse. Acquire the whole U_D, the complete program appendix, and then D1, D2 and D3. For the union K_i of passage owners paid through stage i, the new cost is $c(K_i \setminus K_{i-1})$. Every application is charged in full. Previously acquired prerequisites are not charged again, and no common theory is discounted because another topic uses overlapping arguments.

Added	New cost	Written total	With programs	Prior text used
U_D	15,632	15,632	15,632	—
CODE	27,070	15,632	42,702	—
D1	10,989	26,621	53,691	8,446
D2	13,367	39,988	67,058	3,222
D3	5,499	45,487	72,557	21,663

The last column is the owner union of already acquired text used by the local paper's recorded arguments. It is not a further subtraction from the application's length. The matching historical cumulative costs are 20,916, 41,577 and 47,894. This is a declared reading order, not a research chronology or a claim of a sawtooth pattern.

What a proof route measures. Each of the 37 clauses has a historical route and a current proof or explicit retention/exclusion boundary. The generating-series and faithfulness outputs are shown separately, without new source credit. For a set of prerequisite nodes S, its known text cost is

$$c_{\text{route}}(S) = c\left(\bigcup_{v \in S} \text{owners}(v)\right).$$

Historical arrows document the reasoning asserted in each source; they do not retroactively certify a corrected or overbroad historical lemma. Shared statement passages and program wrappers therefore appear once in a union. Repeated physically printed statements remain counted at every location. Context outside a selected route remains paid in the complete-paper comparisons. No factor is computed from a route subtotal, box sum or number of nodes.

The logical graph has 137 nodes and 284 typed arrows: 50 historical nodes, 58 current nodes, 20 external inputs and 9 verification components. The cost partition has 220 owners and 215 selected-text spans. Every extracted character belongs to exactly one owner in each of the three text policies. The untrimmed sensitivity assigns furniture to explicit owners. Both Unicode normalizations and all five counting units conserve every whole-document total.

Scope and evidence. L066's corrected reversal argument and exceptional full-automorphism computation remain distinct from its historical proof. The Suzuki local normalization, generation, intersection and duality checks are paid alongside the common support count; Suzuki's classification, Lüneburg's structure and other classical inputs retain their original attribution and unknown expanded costs. The Hall–Janko census includes every Borel anchor and the complete geometric recognition tests. Its printed diagrams, full geometric automorphism/correlation groups, doubling relation and supplement annotations remain excluded from the fixed principal census contract. Their source text is still charged.

The five required programs enter logical routes; the three diagnostics have positive printed costs but only validation arrows. The original ten scoped Lean declarations and the later generic supplement of 28 declarations are formal evidence, not complete formalizations of the applications. The latter comprises 12 plus 16 declarations, not 12 in addition to 28.

The input calibration uses the accepted occurrence-bound repairs, including the L066 extracted-character/encoding collision; that embedded font has no ToUnicode stream. No source PDF, primary text or source raster is exported. The offline viewer exposes proof locations, actual edges, owners and unknown inputs. Its deterministic DOM tests are not real-browser layout tests. This ownership and reader packet awaits independent review and changes no accepted paper factor, source credit or question credit.